

● RAG Learning

3.2 Percentages, Growth & Decay

Pearson Edexcel GCSE Mathematics 1MA1 — Ratio, proportion and rates of change — spec refs R9, R16 — Papers 1 (non-calculator), 2 and 3 (calculator)

3.2.1 Percentages as Parts per Hundred (R9)

A **percentage** is a number of parts per hundred: $48\% = \frac{48}{100} = 0.48$.

Percentages above 100% describe amounts larger than the whole: 250% of a quantity is 2.5 times the quantity.

Percentage	Fraction	Decimal
20%	$\frac{1}{5}$	0.2
12.5%	$\frac{1}{8}$	0.125
62.5%	$\frac{5}{8}$	0.625
7.5%	$\frac{3}{40}$	0.075
225%	$\frac{9}{4}$	2.25

Percentages of amounts on the non-calculator paper

Find 10% by dividing by 10, then halve for 5%, halve again for 2.5%, or divide by 100 for 1%. Combine these building blocks.

Worked Example: Work out 22.5% of £280.

$10\% = 28$, so $20\% = 56$; $2.5\% = 7$. Therefore $22.5\% = 56 + 7 = \text{£}63$.

Expressing one quantity as a percentage of another

A as a percentage of $B = \frac{A}{B} \times 100\%$ (convert to the same units first)

Worked Example: (a) 54 minutes as a percentage of 1.5 hours: $\frac{54}{90} \times 100 = 60\%$.
 (b) £21 as a percentage of £14: $\frac{21}{14} \times 100 = 150\%$.

Comparing two quantities using percentages

Worked Example: Mia scored 39 out of 50 on a science test and 57 out of 75 on a history test. Which was the better score?

Science: $\frac{39}{50} = 78\%$. History: $\frac{57}{75} = 76\%$. The **science** score was better.

3.2.2 Percentage Change Using Multipliers (R9)

R9 asks you to interpret percentage changes **multiplicatively**. Instead of working out the change and then adding or subtracting, multiply by a single decimal **multiplier**.

Increase by $r\%$: multiplier = $1 + \frac{r}{100}$ e.g. $+15\% \rightarrow 1.15$

Decrease by $r\%$: multiplier = $1 - \frac{r}{100}$ e.g. $-35\% \rightarrow 0.65$

New amount = original amount $\times$ multiplier

original = 100%

15% $\times 1.15$

65% remains

-35% $\times 0.65$

Increase: multiplier greater than 1. Decrease: multiplier less than 1.

Worked Example: (a) Increase £64 by 15%: $64 \times 1.15 = \text{£}73.60$.(b) A sofa costing £380 is reduced by 35%: $380 \times 0.65 = \text{£}247$.

Finding a percentage change

$$\text{Percentage change} = \frac{\text{actual change}}{\text{original value}} \times 100\%$$

Worked Example: Bus fares rise from £48 to £60 a month. Change = 12. Percentage

$$\text{increase} = \frac{12}{48} \times 100 = \mathbf{25\%}.$$

$$\text{A suitcase's mass falls from 75 kg to 63 kg. Percentage decrease} = \frac{12}{75} \times 100 = \mathbf{16\%}.$$

Exam Tip: Divide by the **original** value. Using the new value ($12 \div 63$) gives 19.0%, which scores no accuracy marks.

Profit and loss

Percentage profit = $\frac{\text{profit}}{\text{cost price}} \times 100$. A market stall buys 60 scarves at £5 each (cost £300), sells 45 at £8 and the other 15 at £3 (takings $360 + 45 = \text{£}405$). Profit = £105, so the percentage profit is $\frac{105}{300} \times 100 = \mathbf{35\%}$.

3.2.3 Original Value Problems (R9)

In an **original value** (reverse percentage) problem you are told the amount *after* a percentage change. Because $\text{new} = \text{original} \times \text{multiplier}$,

$$\text{original} = \frac{\text{new amount}}{\text{multiplier}}$$

Worked Example: After an 8% rise, a season ticket costs £540. What did it cost before?

$$108\% = 540, \text{ so original} = 540 \div 1.08 = \textbf{£500}.$$

Worked Example: In a sale a pair of boots costs £63 after a 30% reduction. Find the original price.

$$70\% = 63, \text{ so original} = 63 \div 0.7 = \textbf{£90}.$$

Worked Example: A bill of £78 includes VAT at 20%. The amount before VAT is $78 \div 1.2 = \textbf{£65}$.

Exam Tip: Do not take 20% of £78 and subtract it: $78 - 15.60 = \text{£}62.40$ is wrong, because the 20% was calculated on the price *before* VAT, not on £78.

Repeated percentage changes

Multiply the multipliers together to find a single overall multiplier.

Worked Example: A share price rises by 20% and then falls by 20%.

Overall multiplier = $1.2 \times 0.8 = 0.96$, so the price ends at 96% of where it started: a **4% decrease**.

Worked Example: Prices are cut by 25%, and then a further 10% is taken off the sale price.

$0.75 \times 0.9 = 0.675$: total reduction = $100\% - 67.5\% = \mathbf{32.5\%}$ (not 35%).

3.2.4 Simple Interest in Financial Mathematics (R9)

With **simple interest**, interest is calculated on the original sum (the principal) only, so exactly the same interest is added each year.

$$\text{Simple interest} = P \times \frac{r}{100} \times n$$

P = principal, r = annual interest rate (%), n = number of years

Worked Example: £1800 is invested for 4 years at 2.5% per year simple interest.

Interest each year = $1800 \times 0.025 = \pounds 45$. Total interest = $45 \times 4 = \pounds \mathbf{180}$.

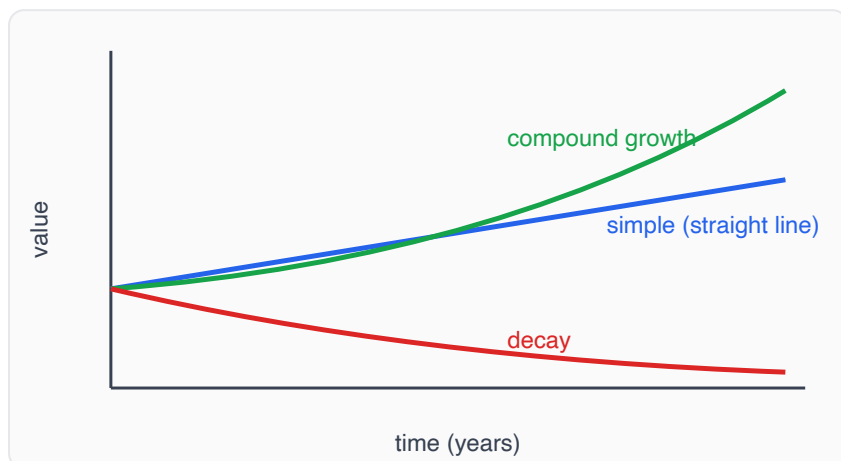
Value after 4 years = $1800 + 180 = \pounds 1980$.

3.2.5 Growth, Decay and Compound Interest (R16)

With **compound interest**, each year's interest is added to the balance and earns interest itself. Population growth, depreciation and radioactive decay follow the same pattern: the **same percentage change** is applied again and again.

$$\text{Value after } n \text{ periods} = P \times \left(1 \pm \frac{r}{100}\right)^n$$

+ for growth or interest, – for decay or depreciation.



Worked Example: £5000 is invested at 3% per year compound interest for 4 years.

$$5000 \times 1.03^4 = 5627.544 \dots \approx \textbf{£5627.54}.$$

Worked Example: A car bought for £22 000 depreciates by 15% each year. Find its value after 3 years.

$$\text{Multiplier} = 0.85. \text{ Value} = 22\,000 \times 0.85^3 = \textbf{£13\,510.75}.$$

Worked Example: A machine costs £30 000. It loses 30% of its value in the first year and 10% in each later year. Value after 3 years:

$$30\,000 \times 0.7 \times 0.9^2 = 30\,000 \times 0.567 = \textbf{£17\,010}.$$

Simple versus compound interest

Worked Example: Account A: £8000 at 5% simple interest for 4 years. Account B: £8000 at 4.8% compound interest for 4 years.

$$A: 8000 + 4 \times 400 = \pounds 9600. \quad B: 8000 \times 1.048^4 = \pounds 9650.17.$$

Account B pays about £50 more despite its lower rate, because the interest compounds.

How many years?

Substitute whole numbers of years into the formula and show the values either side of the target.

Worked Example: A herd of 2500 deer decreases by 6% each year. After how many years will there first be fewer than 2000 deer?

$$2500 \times 0.94^3 = 2076.5 \text{ (still 2000 or more); } 2500 \times 0.94^4 = 1951.9 \text{ (below 2000).}$$

Answer: after **4 years**.

Working backwards

Worked Example: After 3 years of 4% compound interest a balance is £5624.32. Original = $5624.32 \div 1.04^3 = 5624.32 \div 1.124864 = \pounds 5000$.

Worked Example: £6000 grows to £7146.10 in 3 years at $r\%$ per year compound interest. Find r .

$$\left(1 + \frac{r}{100}\right)^3 = \frac{7146.10}{6000} = 1.19101\dots, \text{ so } 1 + \frac{r}{100} = \sqrt[3]{1.19101\dots} = 1.06 \text{ and } r = 6.$$

Exam Tip: "Compound", "each year" and "depreciates by ... per year" all signal a power. Check whether the question wants the final *value* or only the *interest*, and round money to 2 decimal places at the end.

3.2.6 General Iterative Processes (R16) Higher

The Higher version of R16 adds “work with general iterative processes”. These combine a percentage change with a fixed amount added or taken away each period, written as an **iterative formula**:

$$u_{n+1} = a u_n + b$$

u_0 is the starting value, a is the multiplier for the percentage change, and b is the fixed amount added ($b > 0$) or removed ($b < 0$) each period.

Formula	Interpretation
$u_{n+1} = 1.04u_n - 30$	4% growth each period, then 30 removed
$u_{n+1} = 1.025u_n + 200$	2.5% interest a year, then £200 paid in
$P_{n+1} = 0.85P_n + 120$	15% of the population lost, then 120 added
$B_{n+1} = 1.005B_n - 400$	loan at 0.5% interest per month with £400 repaid monthly

Worked Example: $u_{n+1} = 1.04u_n - 30$ and $u_0 = 800$. Find u_2 .

$$u_1 = 1.04 \times 800 - 30 = 832 - 30 = 802$$

$$u_2 = 1.04 \times 802 - 30 = 834.08 - 30 = \mathbf{804.08}$$

Worked Example: A town's population is modelled by $P_{n+1} = 1.1P_n - 60$, with $P_0 = 1000$ in 2025. In which year does the model first give a population above 1500?

$$P_1 = 1040, P_2 = 1084, P_3 = 1132.4, P_4 = 1185.64, P_5 = 1244.20, P_6 = 1308.62, P_7 = 1379.49, P_8 = 1457.44, P_9 = 1543.18$$

It first exceeds 1500 when $n = 9$, i.e. in **2034**.

Long-term behaviour

If the terms settle towards a limit L , then L satisfies $L = aL + b$.

Worked Example: $u_{n+1} = 0.75u_n + 20$. Long-term value: $L = 0.75L + 20 \Rightarrow 0.25L = 20 \Rightarrow L = 80$.

Worked Example: A £6000 loan is charged 0.5% interest per month and only £30 is repaid each month: $B_1 = 1.005 \times 6000 - 30 = 6000$. The monthly interest (£30) equals the repayment, so the balance never goes down.

Exam Tip: On your calculator, enter u_0 and press $=$, then type $1.04 \times \text{Ans} - 30$ and keep pressing $=$. Write down every term you use so the examiner can award method marks, and do not round until the final answer.

Summary

Problem type	Method
Percentage of an amount	Amount $\times$ decimal (or build from 10%, 5%, 1%)
A as a % of B	$\frac{A}{B} \times 100$ (same units)
Increase / decrease	Original $\times$ multiplier
Percentage change	$\frac{\text{change}}{\text{original}} \times 100$
Original value	New amount $\div$ multiplier
Simple interest	$P \times \frac{r}{100} \times n$
Compound growth / decay (R16)	$P \times \text{multiplier}^n$
Iterative processes (R16) Higher	$u_{n+1} = au_n + b$, apply repeatedly

